Jerry Lingjie Mei
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EDUCATION

Princeton University — Princeton, New Jersey
Ph.D. candidate in Computer Science — GPA: 3.85/4.0 — Sept. 2022 - May 2027

Massachusetts Institute of Technology — Cambridge, Massachusetts
M.Eng. in Computer Science — GPA: 4.7/5.0 — Sept. 2020 - May 2022
B.Sc. in Computer Science (6-3), and Mathematics (18) — GPA: 4.9/5.0 — Sept. 2017 - May 2020

Peking University — Beijing, China
B.Sc. candidate in Mathematics — GPA: 3.84/4.00 — Sept. 2016 - Jun. 2017

RESEARCH INTEREST

Procedural world generation for embodied intelligence; 3D vision; generative models.

PUBLICATIONS

** Equal contribution.*

- Alexander Raistrick, Karhan Kayan, Jack Nugent, David Yan, **Lingjie Mei**, Meenal Parakh, Hongyu Wen, Dylan Li, Yiming Zuo, Erich Liang, and Jia Deng. ProcFunc: Function-Oriented Abstractions for Procedural 3D Generation in Python. *arXiv preprint arXiv:2604.26943*, 2026.
- Jack Nugent, Siyang Wu, Zeyu Ma, Beining Han, Meenal Parakh, Abhishek Joshi, **Lingjie Mei**, Alexander Raistrick, Xinyuan Li, and Jia Deng. Evaluating Robustness of Monocular Depth Estimation with Procedural Scene Perturbations. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Alexander Raistrick*, **Lingjie Mei***, Karhan Kayan*, David Yan, Yiming Zuo, Beining Han, Hongyu Wen, Meenal Parakh, Stamatis Alexandropoulos, Lahav Lipson, Zeyu Ma, and Jia Deng. Infinigen Indoors: Photorealistic Indoor Scenes using Procedural Generation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- Shari Liu, Kirsten Lydic, **Lingjie Mei**, and Rebecca Saxe. Violations of physical and psychological expectations in the human adult brain. *Imaging Neuroscience*, 2024; *Proceedings of the 45th Annual Meeting of the Cognitive Science Society (CogSci)*, 2023.
- Alexander Raistrick*, Lahav Lipson*, Zeyu Ma*, **Lingjie Mei***, Mingzhe Wang, Yiming Zuo, Karhan Kayan, Hongyu Wen, Beining Han, Yihan Wang, Alejandro Newell, Hei Law, Ankit Goyal, Kaiyu Yang, and Jia Deng. Infinite Photorealistic Worlds Using Procedural Generation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- **Lingjie Mei***, Jiayuan Mao*, Ziqi Wang, Chuang Gan, and Joshua B. Tenenbaum. FALCON: Fast Visual Concept Learning by Integrating Images, Linguistic descriptions, and Conceptual Relations. *The Tenth International Conference on Learning Representations (ICLR)*, 2022.
- Kevin Smith, **Lingjie Mei**, Shunyu Yao, Jiajun Wu, Elizabeth S. Spelke, Joshua B. Tenenbaum, and Tomer D. Ullman. The fine structure of surprise in intuitive physics: when, why, and how much? *Proceedings of the 42nd Annual Meeting of the Cognitive Science Society (CogSci)*, 2020.
- Kevin Smith, **Lingjie Mei***, Shunyu Yao, Jiajun Wu, Elizabeth S. Spelke, Joshua B. Tenenbaum, and Tomer D. Ullman. Modeling Expectation Violation in Intuitive Physics with Coarse Probabilistic Object Representations. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

SERVICE

- Teaching assistant and preceptor for MIT 6.036, Princeton COS 324, and COS 423.
- Reviewer for NeurIPS 2024, ECCV 2024, NeurIPS 2025, ICCV 2025, CVPR 2026.

RESEARCH SOFTWARE

- princeton-vl/procfunc: coauthor of the library for composing, analyzing, and editing procedural 3D generators in Python.
- princeton-vl/infinigen: core contributor to Princeton VL's procedural scene generator for photorealistic synthetic data.
- ADEPT-Model-Release: model code for the ADEPT intuitive-physics project (NeurIPS 2019).
- ADEPT-Dataset-Release: dataset generation code for ADEPT.
- FALCON-Release: PyTorch reference implementation for the FALCON visual concept learner (ICLR 2022).
- princeton-vl/cvdpack: contributor to tooling for reorganizing/compressing RGB/Depth/Flow/Normal frame datasets.

AWARDS AND ACTIVITIES

- Gold Medal, International Mathematical Olympiad (IMO), 2016.
- Top 25 individual, William Lowell Putnam Mathematical Competition, 2017.
- Boston Dragon Boat Race winner, 2017.
- Member, IEEE-Eta Kappa Nu Honor Society.
- MIT Association of Taiwanese Students, MIT Asian Dance Team, MIT CUCN, and MIT Technique.
- Princeton ECP.